Smart Agricultural Technology 14 (2026) 102214

Contents lists available at ScienceDirect

Smart Agricultural Technology

journal homepage: www.journals.elsevier.com/smart-agricultural-technology

STRep-DETR: a spatio-temporal transformer for efficient detection of sheep behaviours in intensive livestock farm

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ARTICLE INFO

Dataset link: https://github.com/CAU-713/sheepAct

Keywords:
Dense livestock farming
Multi-sheep behavior detection
Spatiotemporal features
Structural Re-parameterization

ABSTRACT

Intensive livestock farming faces significant challenges in monitoring animal behavior, particularly under conditions of medium-to-high stocking densities and changes in light. While computer vision-based detection offers promise, existing methods often struggle with robustness or computational efficiency in these complex real-world settings. To address this, a novel robust model, Detection Transformer with Spatiotemporal Re-parameterization (STRep-DETR), is proposed as an extension of Real-Time Detection Transformer (RT-DETR) for efficient sheep behavior detection. STRep-DETR addresses the aforementioned challenges through three key innovations: (1) Multi-Frame Interaction (MFI) enhancing robustness to density and lighting changes by capturing spatiotemporal cues from reference frames; (2) Dynamic Re-parameterization SpatioTemporal Attention Fusion (DRSTAF) enhances spatiotemporal features from reference frames during training while requiring only the current frame during inference, ensuring both accuracy and efficiency; and (3) A lightweight Pointwise-Group Fusion for Pyramid Network (PGPN) for efficient multi-scale feature integration.

In addition, a sheep behavior detection dataset (SheepAct) is contributed. SheepAct comprises surveillance videos and images capturing 4 representative sheep behaviors (stand, lay, eat, drink) under medium-to-high stocking densities and changes in light. Specifically, STRep-DETR achieves the highest detection accuracy among all compared models for the four representative behaviors, with 85.7% for stand, 89.3% for lay, 98.3% for eat, and 56.2% for drink. At the same time, it maintains an inference speed of 47 frames per second, uses 59 million parameters, and requires 157 billion floating-point operations. This efficiency demonstrates potential for real-time application in intensive livestock monitoring systems.

1. Introduction

Intensive livestock farming has become the dominant model in modern animal husbandry, significantly enhancing production efficiency [1]. However, it also introduces a range of challenges, including concerns over animal health and welfare, as well as complexities in production management. Since animal behavior often reflects underlying physiological states [2], animal behavior monitoring technologies have emerged as promising tools. For example, the feeding behavior of pigs is closely related to their overall health, anomalies such as anorexia or overeating may indicate underlying diseases [3]. Similarly, lameness in dairy cattle has profound effects on milk production and reproductive performance. Early detection is essential for herd management [4]. Monitoring sheep behavior patterns can help identify susceptible or diseased animals at an early stage, enabling timely intervention [5]. Moreover, behavioral data can be integrated into intelligent control systems to aid automated decision-making processes [6]. Effective detection of livestock behavior is a solution that enhances animal welfare, improves resource efficiency, and supports more informed decision-making in intensive livestock systems.

Driven by the rapid development of deep learning and computer vision, significant research efforts have been devoted to advancing animal behavior detection in recent years. Among the widely adopted approaches are one-stage and two-stage object detection frameworks. One-stage detectors, such as YOLO and SSD, perform object classification and localization in a single pass, making them suitable for animal behavior monitoring tasks that require fast inference speed and efficient real-time video processing. A real-time sow behavior detection algorithm was developed using an optimized SSD framework [7], achieving high accuracy in identifying drinking, urination, and mounting

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https://doi.org/10.1016/j.atech.2026.102214
Received 25 November 2025; Received in revised form 12 May 2026; Accepted 12 May 2026
Available online 13 May 2026
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behaviors. PD-YOLO integrates a parallel attention mechanism and depthwise separable convolutions to enhance detection accuracy under crowded conditions while significantly reducing model size and computational complexity, thereby achieving real-time inference speeds suitable for edge deployment [8]. Similarly, FESS-YOLOv8n incorporates FasterNet blocks, selective channel downsampling, and synergistic attention modules to construct a more compact architecture with improved recognition accuracy, demonstrating strong potential for large-scale health monitoring applications [9]. Two-stage detectors, like Faster R-CNN, are favored for their higher detection accuracy, especially in complex environments with occlusions or dense animal populations. To support individual-level behavior monitoring, a Faster R-CNN-based approach was applied for pig identification and head-body association, enabling accurate feeding behavior detection with high precision and recall [10]. A Faster R-CNN-based detector was developed and optimized for poultry feeding behavior detection under changes in light, demonstrating its effectiveness for continuous monitoring in group housing environments [11].

More recently, end-to-end detection frameworks based on transformers have gained attention due to their strong feature modeling capabilities [12]. Several studies have focused on enhancing DEtection TRansformers (DETRs) for animal behavior analysis, such as the RT-DETR-Foalnursing model proposed by [13], which employs SACG-Net as the backbone network to enhance image feature extraction efficiency, enabling more accurate detection of the standing posture of mares and the suckling behavior of foals in stables. Similarly, [14] integrated a dual-layer routing attention mechanism to detect pig behaviors in occluded and overlapping scenarios. Such transformer-based methods excel at capturing long-range dependencies and complex features, improving detection accuracy in difficult scenarios.

Despite significant progress in livestock behavior detection, computer vision-based approaches continue to face challenges posed by high stocking density (i.e., stocking rates greater than 120 kg m^{-2}) and changes in light [4,15]. To address these issues, recent research has increasingly focused on spatiotemporal data-driven methods, which offer enhanced robustness and accuracy in behavior detection [16,17]. A skeletal representation framework combining YOLOv5, Deep-SORT, and AlphaPose was developed for sheep behavior detection [18]. Another approach utilizes Two-Stream Convolutional Networks to extract discriminative spatiotemporal features from video sequences for porcine behavior analysis [19]. To mitigate the adverse effects of changes in light on behavior detection, the DB-TEAF method incorporates a temporal excitation and aggregation mechanism with frequency channel attention to emphasize key spatiotemporal features under different illumination environments [20]. Nonetheless, these approaches often import computing resources, making them less suitable for real-time farm monitoring systems.

To address these challenges, STRep-DETR is designed for practical sheep behavior detection in real farm environments. During training, STRep-DETR utilizes multiple consecutive frames to capture spatiotemporal dependencies that are crucial for behavior understanding. Through spatiotemporal re-parameterization, these temporal representations are compressed, enabling the model to perform fast inference using a single frame while maintaining the benefits of temporal learning. This design achieves a balance between efficiency and accuracy, making STRep-DETR for livestock behavior monitoring. The main contributions of our work can be summarized as follows:

(1) A Multi-Frame Interaction (MFI) module is introduced to enhance robustness under medium-to-high stocking densities and changes in light by effectively capturing spatiotemporal cues from reference frames.
(2) A novel Dynamic Re-parameterization SpatioTemporal Attention Fusion (DRSTAF) module is designed, which leverages both current and reference frames during training while requiring only the current frame during inference, thereby improving detection efficiency without compromising accuracy.
(3) A lightweight feature fusion module, Pointwise-Group Fusion for Pyramid Network (PGPN), is proposed to efficiently integrate spatiotemporal and multi-scale appearance features, striking a balance between performance and computational cost.
(4) We construct a realistic multi-behavior sheep dataset, SheepAct, which captures four typical behaviors under varying illumination and stocking densities, providing consecutive frames for effective spatiotemporal behavior detection.

2. Materials and methods

In this section, the materials and methods used to develop and evaluate STRep-DETR are detailed. This includes the construction of the sheep behavior dataset and the design of the proposed model architecture. By systematically outlining the dataset and model components, a clear and reproducible foundation is provided for understanding how effective spatiotemporal behavior detection is achieved in intensive livestock farming through STRep-DETR.

2.1. Materials

2.1.1. Experimental materials and data acquisition

Surveillance video was collected at Tianjin Aoqun Animal Husbandry Co., Ltd., China, between July 2023 and January 2024. All monitored pens shared identical structural features: each was enclosed by fences to prevent sheep from escaping and included a front-positioned feeding trough. Videos were recorded from 4 different sheep pens under medium-to-high stocking densities, using Dahua 1230C-A cameras (1920 × 1080 resolution, 25 FPS). A total of 123 continuous 24-hour video recordings were obtained from each of the 4 pens, resulting in 496 videos in total. The collected videos spanned all times of day, capturing significant variations in lighting conditions. As shown in Fig. 1, the surveillance footage includes representative scenes from 4 sheep pens under varying illumination.

Each pen covered approximately 24 m^2, with concrete flooring partially overlaid by blue rubber mats to enhance traction. Each pen housed approximately 10 to 34 sheep, corresponding to a stocking density of about 0.4 to 1.4 animals per m^2. The background included fixed structures such as fences, troughs, and sidewalls, as well as moderate clutter from feed residues and bedding materials, adding realistic scene complexity. Illumination conditions ranged from natural daylight to artificial LED lighting at night, producing color temperatures between roughly 3000 K and 6000 K. This broad spectral variation provides a realistic foundation for evaluating the visual adaptability of detection models in practical farm environments.

2.1.2. Dataset construct and online augmentation

Existing public datasets for livestock behavior analysis, such as CBVD-5 [21] and DairyGoat [5], primarily focus on individual identification, posture recognition under controlled illumination, or coarse-grained action classification. However, these datasets exhibit several limitations for real-world spatiotemporal behavior detection in intensive farming conditions: (1) most are collected under low-density or individually monitored settings, lacking the frequent occlusions and social interactions that occur in group-housed environments, and (2) temporal context is typically ignored, as many datasets consist of isolated images rather than continuous video sequences. To address these limitations, we constructed a new behavior detection dataset, SheepAct, designed to capture realistic and complex conditions in multi-animal environments. During dataset preparation, low-quality video clips and images with insufficient visual information were first filtered out to ensure data reliability. Specifically, videos were excluded if the camera was out of focus, if the lens was partially obstructed by dust or spider webs, or if illumination caused severe overexposure or underexposure,

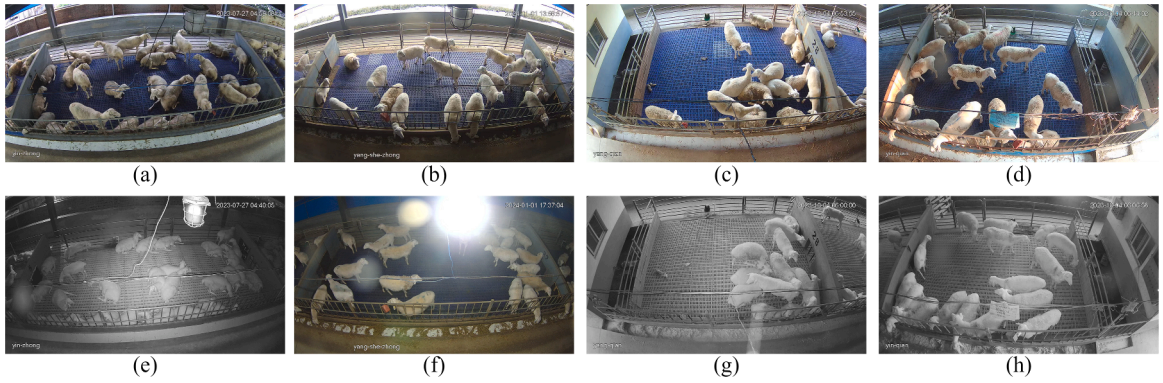


Fig. 1. Representative images from 4 sheep pens under daytime and nighttime conditions. (a) and (e) are from sheep pens I, (b) and (f) are from sheep pens II, (c) and (g) are from sheep pens III, and (d) and (h) are from sheep pens IV. The top row (a–d) corresponds to daytime scenarios, and the bottom row (e–h) to nighttime ones.

making the sheep's body contour indistinct. Each video was manually inspected, and clips meeting any of these conditions for more than 20% of their duration were discarded. After filtering, 468 high-quality 24-hour surveillance videos were retained, each capturing diverse sheep behaviors across varying lighting and density conditions.

For dataset construction, each 24-hour video was divided into 2-hour segments to capture behavioral variations across the day. From each segment, one 12-second video clip was randomly selected without temporal overlap, avoiding redundancy and ensuring coverage of different time periods. Each 12-second clip contained 300 consecutive frames recorded at 25 FPS. To build spatiotemporal samples, five frames were extracted at 3-second intervals, including the current frame and four preceding frames, forming a compact motion sequence that preserved short-term dynamics and contextual information. This process yielded 234 annotated video clips (12 seconds each), containing a total of 5,137 labeled sheep for model training. For evaluation, each 24-hour video was divided into 10-hour segments, and representative frames were uniformly sampled at 30-minute intervals to construct the test set. This resulted in 48 images with 515 annotated sheep. Overall, the dataset maintained an approximate 9:1 train–test split, providing sufficient diversity for model learning while enabling reliable assessment of generalization performance.

SheepAct includes 4 specific behaviors (stand, lay, eat, and drink), which were selected for their strong relevance to monitoring animal health and welfare. Table 1 describes these behaviors in detail. Fig. 2 depicts an example of these 4 sheep behaviors. We use the open-source tool X-AnyLabeling to create samples and store them in the widely used COCO format to save annotations, with structured fields including image metadata, behavior categories, and bounding box coordinates. All annotations were performed by a single annotator with professional experience in animal husbandry and prior training in computer vision data labeling. To ensure labeling consistency, the annotator followed a predefined annotation protocol with detailed behavioral definitions and visual reference examples. During the labeling process, ambiguous cases were flagged and reviewed multiple times to maintain consistency across the dataset. Additionally, a second annotator with a professional background in animal husbandry independently verified 35% of the training set samples. The inter-rater agreement for behaviors achieved a Cohen's Kappa coefficient greater than 0.95, indicating low annotator bias risk.

Table 1
The specific explanations of the behaviors in this study.

Behavior	Description of behavior
Stand	Legs are straight and supporting the body
Lay	Legs are bent and sitting on the floor
Eat	Standing with the head in the feed tray
Drink	Standing with the head in the water troughs

In SheepAct, annotations are performed on the last frame of each video clip, referred to as the current frame, and are also applied to images. Samples are categorized into two types: training samples, which support spatiotemporal context in behavior detection, and inference samples, which are used to evaluate model performance based solely on the current frame. Specifically, each training sample includes annotation along with a 12-second video clip comprising both the current frame and reference frames. In contrast, each inference sample consists of annotation and a image representing the current frame. Fig. 3 illustrates statistical characteristics of SheepAct. Fig. 3a shows the frequency distribution of annotated sheep behaviors. Fig. 3b presents the frequency distribution of the number of sheep present in each sample, providing an overview of group sizes encountered in SheepAct.

To improve model generalization and robustness, online data augmentation was employed during training. Unlike offline augmentation, which generates and stores transformed data in advance, online augmentation applies transformations on-the-fly during each training epoch, thereby increasing data diversity without requiring additional storage. Specifically, only basic data augmentations were applied, including random color distortion, expansion, cropping, flipping, and resizing operations, follow [22].

2.2. *RT-DETR network model*

The Real-Time Detection Transformer (RT-DETR) is an efficient end-to-end object detector [22]. It discards the traditional region-proposal pipeline in favor of a fully integrated architecture, where a hybrid encoder handles multi-scale representations and an efficient decoder directly predicts object queries. This design removes hand-crafted components such as non-maximum suppression and anchor generation, resulting in a cleaner optimization process and more stable training dynamics. By jointly capturing global context and local structural cues, RT-DETR retains the expressive capability of transformer-based detectors while notably reducing computational cost.

RT-DETR employs a hybrid encoder that enhances multi-scale features through Attention-based Intra-scale Feature Interaction (AIFI) and CNN-based Cross-scale Feature Fusion (CCFF), producing enriched feature sequences for subsequent decoding. It further applies Uncertainty-minimal Query Selection to identify a fixed set of representative features as initial object queries. These queries are progressively refined by the decoder through an auxiliary prediction head until final bounding boxes and confidence scores are produced.

With these components, RT-DETR achieves a favorable trade-off between accuracy and throughput on mainstream benchmarks. Its strong detection performance, streamlined end-to-end pipeline, and hardware-efficient design make it a practical choice for applications requiring both precision and speed, including autonomous driving, retail analytics, and edge-side perception. However, behavior detection

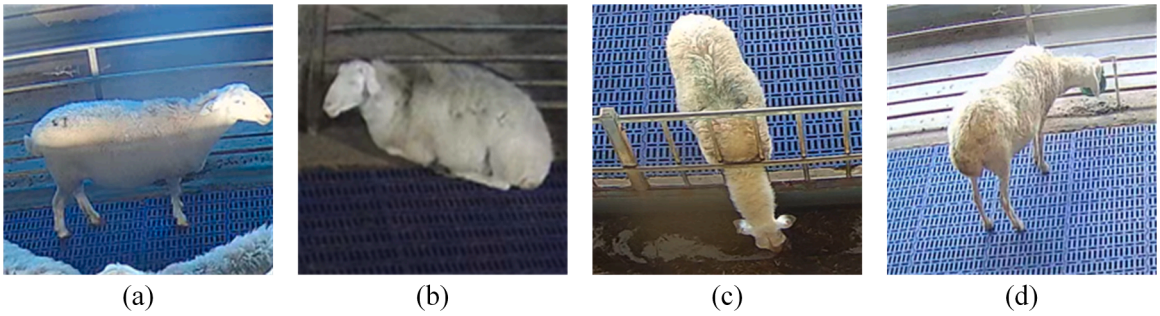


Fig. 2. Different behaviors of sheep. (a), (b), (c) and (d) respectively represent stand, lay, eat and drink.

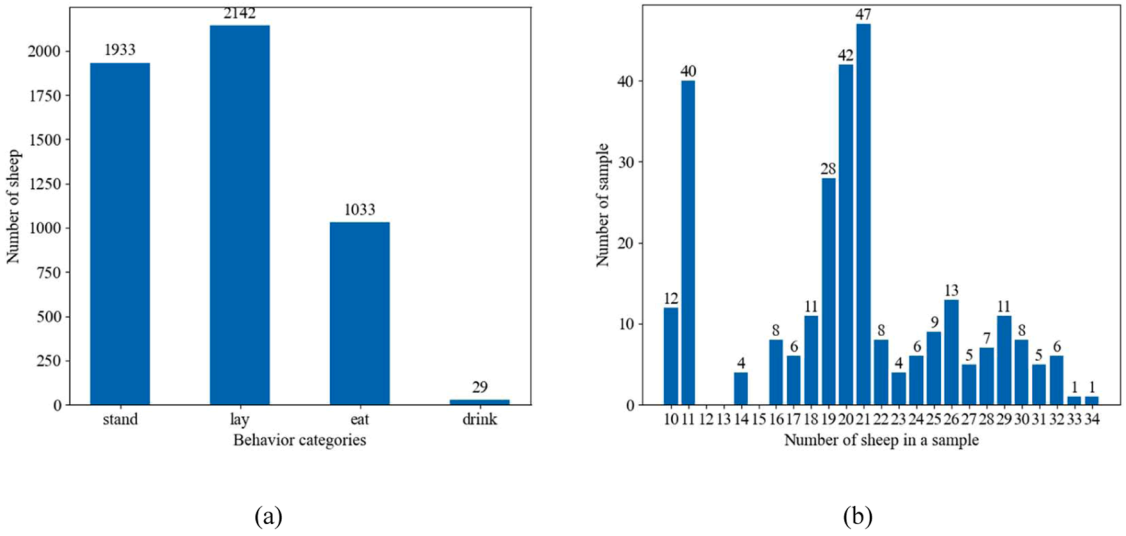


Fig. 3. Statistics of sheep behavior and density in SheepAct. (a) Distribution of different behavioral categories (b) Distribution of sheep density.

tasks often demand richer spatiotemporal representations and explicit modeling of interactions between objects. RT-DETR, like other DETR-based frameworks, lacks such mechanisms, which limits its suitability for fine-grained tasks such as sheep behavior analysis.

2.3. Design of the STRep-DETR model architecture

2.3.1. Overall Architecture

To address the challenges of accurately detecting sheep behaviors under medium-to-high stocking densities and changes in light while maintaining real-time performance, STRep-DETR was proposed, built

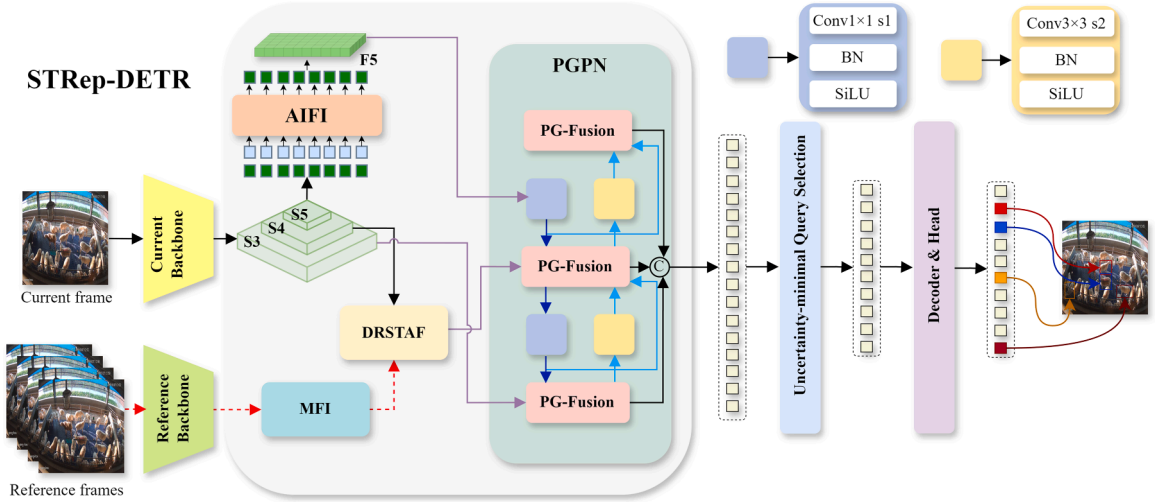


Fig. 4. Overall architecture of STRep-DETR. MFI, DRSTAF, and PGPN are the modules proposed in this study. Current Backbone and Reference Backbone use ResNet-50 pre-trained on ImageNet [23]. AIFI denotes encoder, Uncertainty-minimal Query Selection proposed in RT-DETR [22]. Decoder&Head uses DINO header [24]. BN denotes batch normalization layer, SiLU is the SiLU activation function. The symbol "C" denotes concat.

upon the RT-DETR framework. It inherits many of RT-DETR's advantages, such as fast convergence and efficient multi-scale feature interaction. The overall architecture of STRep-DETR is illustrated in Fig. 4. STRep-DETR incorporates the following key components: Current Backbone and Reference Backbone for feature extraction, AIFI encoder for feature enhancement, MFI for efficient spatiotemporal feature extraction, DRSTAF to accommodate different data pipelines, PGPN for efficient feature fusion, Uncertainty-minimal Query Selection to initialize decoder queries, and a Decoder with Head for bounding box prediction.

The data pipeline of STRep-DETR is illustrated in Algorithm 1 and consists of distinct procedures for training and inference modes. In training mode, STRep-DETR takes the current frame and reference frames as input. Current Backbone extracts features from current frame and enhances them via AIFI. Reference Backbone extracts features from reference frames and captures spatiotemporal representations via MFI. DRSTAF effectively fuses current frame features and spatiotemporal features. Subsequently, PGPN performs multi-scale feature aggregation, generating a coherent sequence of features. Then, Uncertainty-minimal Query Selection initializes decoder queries, which are subsequently processed by the Decoder and Head to produce bounding boxes and confidence scores. During inference mode, only the current frame is processed, where features are enhanced via DRSTAF before undergoing subsequent processing.

2.3.2. Multi-Frame Interaction (MFI)

Accurately detecting sheep behaviors from video data is difficult in dense farm environments because frequent occlusions, visual similarity among individuals, and subtle posture transitions reduce the discriminative information available in single frames. While 3D convolution-based approaches can capture short-term motion cues, they usually require substantial computational cost and struggle to model long-range temporal relationships efficiently [25,26]. Recent studies show that attention-based spatiotemporal modeling offers a more flexible alternative by enabling the network to selectively emphasize informative spatial regions and temporally salient frames [27,28].

Motivated by these observations, we develop MFI, whose overall structure is shown in Fig. 5, to aggregate complementary information from multiple reference frames through an attention-driven mechanism that adaptively emphasizes spatially salient regions and temporally informative cues. This approach is well suited to dense farm environments where frequent occlusions and subtle posture transitions limit the discriminative power of single-frame features. MFI adopts a concise two-stage design in which local extraction enhances intra-frame representations and global interaction captures cross-frame temporal dependencies, providing an efficient and coherent means of integrating spatial detail with temporal continuity for robust behavior discrimination.

Local extraction sub-module composed of Depthwise Convolution (DWConv) and Multi-Head Self-Attention (MHSA) is utilized. DWConv captures intra-channel spatial patterns, while MHSA models long-range dependencies across spatial locations and feature dimensions. These operations are applied to feature maps extracted from 4 reference frames. Global interaction sub-module composed of pointwise convolution (PWConv) and MHSA is used to capture inter-frame correlations. PWConv facilitates inter-channel feature fusion by adaptively integrating representations across frames, thereby strengthening cross-frame contextual relationships. Then, MHSA models global spatiotemporal dependencies by dynamically attending to informative patterns across spatial locations and temporal dimensions via learnable attention weights. BN is applied at the end of each sub-module to stabilize training and normalize inter-frame feature distributions.

Algorithm 1
Data pipeline of STRep-DETR In training and inference mode.

```
# CF: Current frame
# RF: Reference frames
# CB: Current Backbone
# RB: Reference Backbone
# S3, S4, S5: Last three stage outputs of CB
# F5: AIFI enhanced feature from S5
# FRF: Feature Reference frames
# O: Output of DRSTAF

# Extract features of CF from CB
S3, S4, S5 = CB(CF)
F5 = AIFI(S5)

if mode == training:
# In training mode, process each F of RF through RB
for F in RF do:
FRF.append(RB(F))
end for
# Extracting spatiotemporal feature from RF
spatiotemporal_feature = MFI(FRF)
# Fuse CF and RF features
O = DRSTAF(S4, spatiotemporal_feature)
else:
# In inference mode, only CF is used
O = DRSTAF(S4)

# Generate boxes and confidence scores
out = Uncertainty-minimal Query Selection(PGPN(S3, O, F5))
out = Head(Decoder(out))
```

2.3.3. Dynamic Re-parameterization SpatioTemporal Attention Fusion (DRSTAF)

To enhance spatiotemporal feature modeling while maintaining computational efficiency during inference, we design DRSTAF, a novel module that achieves training–inference decoupling through structural re-parameterization. The core innovation of DRSTAF lies in its ability to leverage both the current feature map and reference feature map during training to learn rich spatiotemporal correlations but require only current feature map at inference time. This is made possible by our proposed Re-parameterized Current-Reference Fusion (RCRF) component, which fuses multi-frame information into a single-path convolutional operator via kernel-level merging.

The motivation for this design stems directly from the practical inference constraint in livestock monitoring: real-time inference must rely on a single frame, yet behavior understanding benefits from multi-frame context. RCRF resolves this tension with negligible approximation error, as validated in Table 9.

As illustrated in Fig. 6, the core idea of RCRF is to leverage spatiotemporal correlation between the current frame and reference frames during training, while relying solely on the current frame during inference. This is achieved through structural re-parameterization, a technique that merges multi-branch training-time modules into a single equivalent path at inference time, as pioneered in Diverse Branch Block [29].

RCRF employs two convolutional layers to process the current feature map I and the reference feature map V separately. During inference, these are fused via re-parameterization, allowing the model to approximate the same output using only I. The formulation of RCRF is expressed as *Eq. 1*.

$$O = \begin{cases} \mathrm{Conv}_{3\times 3}(I) + \mathrm{Conv}_{1\times 1}(V), & \text{if mode} = \text{training} \\ \mathrm{ReConv}(I), & \text{if mode} = \text{inference} \end{cases} \tag{1}$$

where I and V denote the current and reference feature maps of identical spatial and channel dimensions, O represents the fused feature output, and *ReConv* represents the operation performed during inference using the pre-fused convolutional kernels from $Conv_{3\times 3}$ and $Conv_{1\times 1}$.

In practice, we construct a *ReConv* with kernel F' and bias b', where these parameters are derived from the trained $Conv_{3\times 3}$ and $Conv_{1\times 1}$ layers. The fusion process can be expressed as *Eq. 2*.

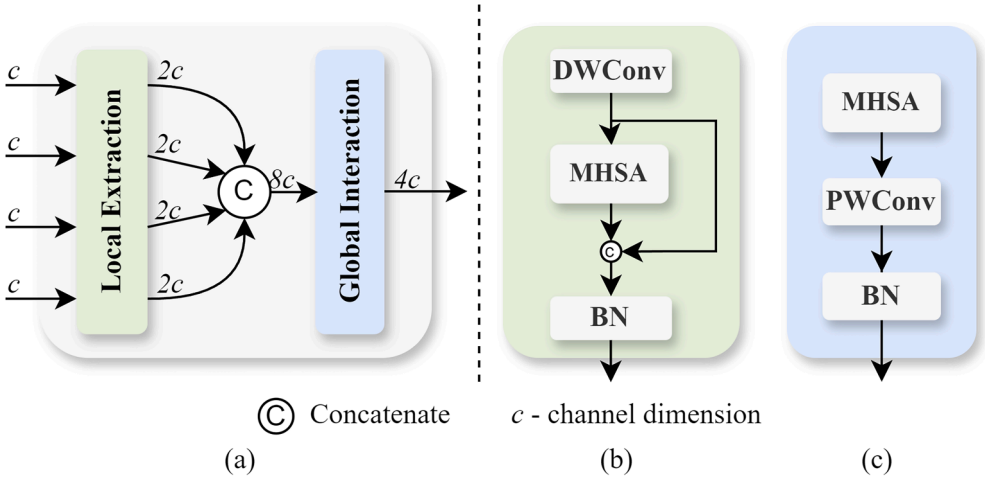


Fig. 5. Illustration of MFI: (a) overall architecture, (b) Local Extraction sub-module, and (c) Global Interaction sub-module.

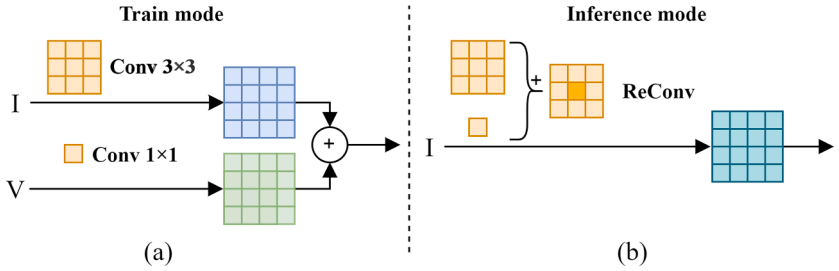


Fig. 6. Structure of RCRF: (a) training mode and (b) inference mode. The symbol "+" denotes element-wise addition.

$$F' = F^{3\times3} + F^{1\times1}, b' = b^{3\times3} + b^{1\times1} \quad (2)$$

where $F^{3\times3}$ and $F^{1\times1}$ represent the equivalent kernels obtained from $Conv_{3\times3}$ and $Conv_{1\times1}$ after appropriate padding and alignment, respectively, and $b^{3\times3}$ and $b^{1\times1}$ are their corresponding biases. To enable element-wise addition between the 3 × 3 and 1 × 1 convolutional outputs during re-parameterization, the 1 × 1 kernel is zero-padded to 3 × 3 prior to fusion.

However, while RCRF enables efficient temporal compression, its static fusion kernel cannot adapt to complex, non-rigid motion patterns, such as rapid head lowering during drinking. To address this without compromising the re-parameterizable structure, we further embed deformable convolutions within the RCRF output path. As shown in Fig. 7, the fused feature is processed by three parallel deformable convolutions to capture multi-scale motion offsets, followed by pointwise convolution and residual connection. Critically, these deformable layers operate solely on the re-parameterized output, preserving the single-frame inference behavior while enhancing feature adaptability.

2.3.4. Pointwise-Group Fusion for Pyramid Network (PGPN)

In practical agricultural settings, computational resources are often limited, necessitating lightweight yet powerful feature fusion architectures. While conventional feature pyramids such as ASFF [30] and BiFPN [31] achieve strong performance in general object detection, they incur FLOP costs due to dense cross-scale connections. To address this, we propose PGPN, a novel lightweight fusion module that combines the efficiency of group convolution [32–34] with the representational flexibility of structural re-parameterization [35].

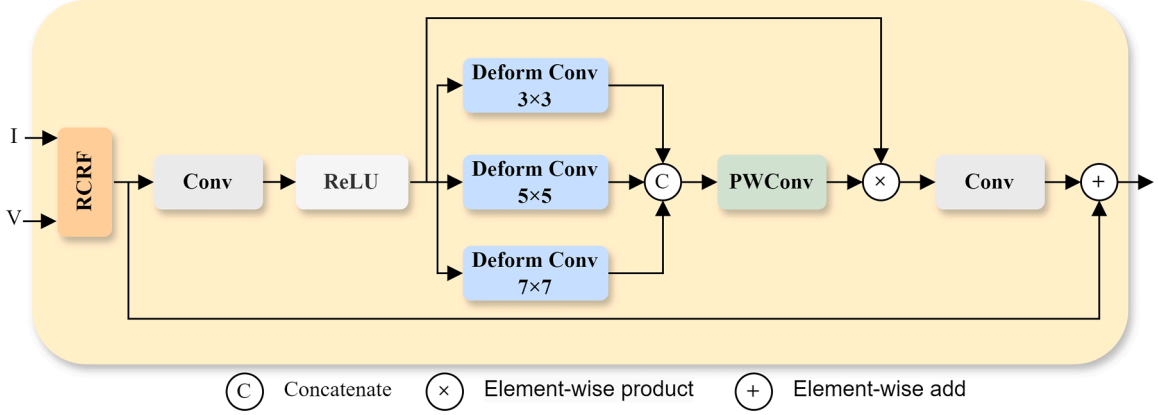


Fig. 7. Pipeline of DRSTAF.

PGPN is constructed around a core operator termed Pointwise Group Convolution (PGConv). PGConv partitions channels into groups so that computation is reduced while partial inter channel communication is preserved, which allows it to maintain stronger representational capacity than depthwise separable convolution at a significantly lower cost than standard pointwise convolution. Let the input feature map is given by $X \in R^{H\times W\times C_{in}}$, and the output feature map is represented as $Y \in R^{H\times W\times C_{out}}$ where H and W denote the height and width of the feature maps, respectively, while C_{in} and C_{out} represent the number of input and output channels. PGconv divides both input and output channels into G groups, which can be formulated as follows *Eq. 3*.

$$Y_{h,w,c_{out}} = \sum_{g=0}^{G-1} \sum_{c_{in}=0}^{C_{in}/G-1} X_{h,w,g\cdot C_{in}/G+c_{in}} \cdot K_{g,c_{in},c_{out}} \tag{3}$$

where g is the group index. c_{in} denotes the input channel index within the group, while c_{out} represents the output channel index. $K_{g,c_{in},c_{out}}$ corresponds to the 1×1 convolution kernel within group g. When $G = c_{in}/4$, which provides a practical balance between computational efficiency and intra group feature interaction.

To further improve the efficiency of feature fusion, we propose Pointwise Group Fusion (PG-fusion), a novel fusion module that alternates feature processing between PGconv and RepBlock, as illustrated in Fig. 8. Here, RepBlock refers to a re-parameterizable convolutional block introduced in prior work, which is designed to improve inference efficiency by enabling the fusion of multi-branch structures into a single-path architecture at inference time [35]. PG-fusion achieves more efficient feature interaction by alternating channel-wise processing and structural Re-parameterization, thereby avoiding redundant computations inherent in traditional approaches.

Building on PG-fusion, we further develop the PGPN in Fig. 4, which extends the PGConv across multiple scales within a hierarchical structure. PGPN significantly boosts computational efficiency and representation capability by enabling fine-grained feature fusion across different resolutions.

3. Results

3.1. Experiment detail and evaluation metric

The detailed experimental environment configuration is shown in Table 2, and training and inference hyperparameters are summarized in Table 3. The training and inference processes of all models in this study were run on the same experimental setup. To ensure comparability and reproducibility, all hyperparameter settings strictly follow those used in RT-DETR [22].

In experimental evaluation, we adopted $AP_{50}(\%)$ as the primary metric to assess the performance of the proposed behavior detection method. The detailed computational procedures are presented in *Eq. 4*, *Eq. 5*, and *Eq. 6*.

$$P = \frac{TP}{TP + FP} \times 100\% \tag{4}$$

$$R = \frac{TP}{TP + FN} \times 100\% \tag{5}$$

Table 2
Experimental environment configuration.

Component	Version
GPU	NVIDIA GeForce GTX RTX 4090
Operating System	Ubuntu22.04
Programming language	Python 3.12
Framework	PyTorch 2.3.0
Cuda	12.1

Table 3
Training and inference hyperparameters.

Hyperparameter	Value
Image input	640px * 640px
Training batch size	8
Inference batch size	1
Initial learning rate	0.0001
Optimizer algorithm	AdamW with a weight_decay of 0.0001
Optimizer β_1	0.9
Optimizer β_2	0.999
Epoch	120

$$AP = \int_0^1 P(R)dR \tag{6}$$

Here, Precision (P) refers to the proportion of correctly identified positive samples among all predicted positives, reflecting the model's detection accuracy. Recall (R) denotes the ratio of correctly detected true positives to all actual positive instances, indicating detection coverage. In these formulas: True Positives (TP) representing correctly identified behavior instances, False Positives (FP) denoting instances misclassified as other behaviors, and False Negatives (FN) indicating undetected true behaviors. In this study, AP with a threshold of 0.5 (denoted by $AP_{50}(\%)$) was adopted because it balances localization accuracy and computational efficiency, which aligns with our real-world inference constraints.

To analyze model performance under different occlusion conditions, we define the occlusion ratio (OR) using a standard overlap-based occlusion estimation strategy to quantify the extent to which a target instance is occluded by other instances. Specifically, the occlusion ratio for the i-th instance is defined as *Eq. 7*.

$$OR_i = \frac{|O_i|}{|B_i|} \tag{7}$$

where B_i denotes the area of the target bounding box, and O_i represents the union area of all overlapping regions between the target box and other instance boxes in the same frame.

To comprehensively evaluate model efficiency, we additionally analyzed computational complexity through three metrics: Floating-Point Operations (FLOPs) for operation count, parameter quantity (Params) for model size, and inference speed in Frames Per Second (FPS) for real-time processing capability. The formulas of FLOPs, Params and FPS are shown in *Eq. 8*, *Eq. 9*, and *Eq. 10*.

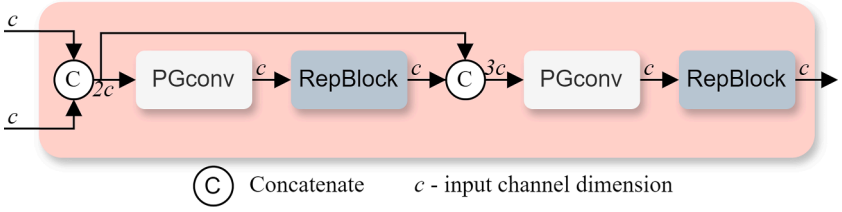


Fig. 8. Architecture of PG-Fusion.

$$FLOPs = \sum \left(2 \times K^2 \times C_{in} \times C_{out} \times H \times W\right) \tag{8}$$

$$Params = \sum \left(\left(K^2 \times C_{in} + 1\right) \times C_{out}\right) \tag{9}$$

$$FPS = \frac{FrameNum}{ElapsedTime} \tag{10}$$

Where K is the size of the convolution kernel, C_{in} is the number of input channels, C_{out} is the number of output channels, and H and W represent the height and width of the output feature map, respectively. *FrameNum* denotes the total number of input frames processed. Additionally. *ElapsedTime* measures the total execution duration of the operation. In this study, FPS was measured under a batch size of 1 and is denoted as $FPS_{bs=1}$.

3.2. Comparison with SOTA methods

To evaluate the overall performance of STRep-DETR, we compare it with several representative state-of-the-art detectors, including the recent YOLO series, namely YOLOv11 [36], YOLOv12 [37], and YOLOv13 [38], as well as four advanced DETR-based architectures: Conditional-DETR [39], DAB-DETR [40], DINO [24], RT-DETR [22] and RT-DETRv2 [41]. At the same time, to balance performance and detection efficiency, we selected the YOLO-X variants for comparison. These models were selected because they represent the two main paradigms of modern object detection: anchor-based convolutional frameworks and transformer-based end-to-end architectures. It is worth noting that, STRep-DETR leverages temporal context during training to improve feature representation, the model operates strictly on single-frame inputs during inference, without requiring any additional temporal information. Therefore, the comparison reflects a reasonable evaluation in terms of input modality and computational cost. The results obtained on the SheepAct dataset are summarized in Table 4.

We conducted 5 independent training runs with different random seeds for both STRep-DETR and RT-DETR. Table 4 reports the mean and standard deviation over these runs. The results show that both models exhibit stable performance across most categories. In particular, STRep-DETR achieves low variance on major classes such as Eat (98.3 ± 0.8) and Lay (89.3 ± 0.8), indicating consistent training behavior despite the small training set. As expected, the Drink category shows relatively higher variance (56.2 ± 3.9) due to its extremely limited number of training samples. Nevertheless, STRep-DETR consistently outperforms RT-DETR (50.6 ± 3.1) across all runs, demonstrating improved robustness. Importantly, the performance gains of STRep-DETR over RT-DETR are consistent across all categories, with improvements of +4.7 (Eat), +6.1 (Lay), +9.0 (Stand), and +5.6 (Drink) in AP_{50}. These consistent margins, together with relatively small standard deviations, indicate that the observed improvements are statistically reliable rather than arising from random fluctuations.

In terms of overall performance, STRep-DETR achieves a favorable trade-off between detection accuracy and computational efficiency among all compared models. As shown in Table 4, it delivers the highest AP_{50} scores across all behavioral categories, outperforming both transformer-based and CNN-based detectors. At the same time, STRep-DETR maintains competitive inference speed and moderate computational cost, achieving 47 FPS with 157 GFLOPs, outperforming DINO and DAB-DETR in speed while remaining comparable to RT-DETR. These results establish a strong performance foundation, which is further analyzed through detailed ablation studies in the following section.

3.3. Ablation study

3.3.1. Component contribution analysis

To analyze the contribution of each component, we conduct an ablation study based on the RT-DETR baseline. The analysis follows a stepwise approach, where MFI, DRSTAF, and PGPN are incrementally introduced to isolate their individual and combined effects. The results are summarized in Table 5.

Introducing MFI to the RT-DETR baseline yields consistent performance improvements across all behavior classes, with notable gains in Stand (+6.3%) and Eat (+2.0%). The MFI & concat configuration fuses spatiotemporal features via direct point-wise concatenation with image features, effectively enhancing temporal continuity. This experiment provides a controlled reference for subsequent module integration, clarifying that feature-level fusion alone contributes to improved detection.

Replacing concat with DRSTAF further enhances accuracy, particularly for Lay (+2.4%) and Drink (+5.8%), by introducing adaptive attention weights that dynamically align cross-frame motion cues. Although this refinement slightly increases the computational load (from 157G to 174G FLOPs), the inference speed remains practical at 44 FPS, confirming an efficient trade-off between accuracy and complexity.

Incorporating PGPN into the baseline independently yields consistent performance improvements, particularly for Eat (+0.8%) and Lay (+3.0%). The module aggregates multi-scale spatial features, effectively preserving representation capacity while minimizing computational redundancy. Importantly, PGPN achieves these accuracy gains alongside a substantial reduction in model complexity, decreasing parameter count by 11.4% and FLOPs by 12.4%, reflecting its lightweight architectural design.

Collectively, these results demonstrate the complementary roles of the three proposed modules within a unified spatiotemporal representation framework. MFI establishes temporal continuity by extracting and aligning multi frame features across behavioral sequences. DRSTAF refines these representations through adaptive attention, preserving discriminative motion cues while maintaining computational tractability. PGPN focuses on lightweight feature aggregation: it reduces parameter count and computational cost while delivering consistent, albeit modest, accuracy gains across behavior classes. When jointly deployed, these components operate synergistically, enabling STRep-DETR to balance detection accuracy, model compactness, and inference efficiency. The ablation study confirms that each module delivers

Table 4
Model comparison result.

Model	AP_{50} (%)				Params(M)	$FPS_{bs=1}$	FLOPs(G)
	Eat	Lay	Stand	Drink			
YOLOv11 X	94.4	86.5	78.7	52.7	56.9	29	194.9
YOLOv12 X	96.8	78.1	73.8	50.7	59.1	31	199.0
YOLOv13 X	97.2	80.5	72.0	49.0	64.0	27	199.2
Conditional-DETR	57.1	26.6	15.2	50.5	**43**	29	195
DAB-DETR	43.5	24.6	15.9	13.4	43	29	195
DINO	78.4	79.4	57.3	52.7	47	23	279
RT-DETR	93.6±0.7	83.2±1.9	76.7±2.0	50.6±3.1	44	**49**	**137**
RT-DETRv2	93.4	84.7	78.4	51.3	44	49	137
STRep-DETR (our)	**98.3±0.8**	**89.3±0.8**	**85.7±1.7**	**56.2±3.9**	59	47	157

Table 5
Contribution of different modules to network.

Model	AP_{50} (%)				Params(M)	$FPS_{bs=1}$	FLOPs(G)
	Eat	Lay	Stand	Drink			
RT-DETR(baseline)	93.2	83.4	77.3	50.9	44	49	137
+ MFI&concat	95.2	84.1	83.6	51.7	73	47	157
+ MFI&DRSTAF	95.9	86.6	84.6	56.3	64	44	174
+ PGPN	94.0	86.4	77.3	52.6	**39**	**51**	**120**
STRep-DETR(our)	**97.7**	**90.1**	**85.7**	**58.7**	59	47	157

measurable gains in isolation, and their combination yields optimal performance without introducing disproportionate computational overhead. This validation supports a complementary integration strategy that effectively addresses the distinct challenges of behavior recognition.

3.3.2. *Spatiotemporal feature extraction comparison and evaluation*

To evaluate the trade-off between detection performance and computational efficiency of the proposed MFI, we conducted comparative experiments on several spatiotemporal feature extractors, including C3D [25], X3D [42], Video Swin Transformer (VideoSwin) [43], and Spatio-Temporal Action TransformeR (STAR) [44]. These extractors were selected to comprehensively cover the main paradigms of video feature modeling: 3D convolutional networks (C3D, X3D) and attention-based transformer frameworks (VideoSwin, STAR). In particular, STAR represents a recent state-of-the-art transformer-based model, which provides a strong and up-to-date baseline for comparison. Detailed experimental results are shown in Table 6.

As shown in Table 6, MFI shows a balance between detection accuracy and computational efficiency, yielding the highest FPS and the lowest FLOPs among all compared methods. Although STAR achieves slightly higher AP_{50} on Eat, Lay, and Stand, these gains come at a substantially higher computational cost, requiring nearly three times the FLOPs of MFI. In addition to the overall efficiency advantage, MFI achieves the largest category-level improvement on Drink (+5.0% compared to STAR), while maintaining competitive performance on the remaining categories.

To better understand the behavior of MFI under challenging scenarios, further analyses are conducted from the perspectives of occlusion and confusion. Test instances are grouped according to OR, and the AP_{50} results under different occlusion ranges are reported in Table 7.

MFI consistently outperforms the baseline across all occlusion ranges, with the most notable improvement observed in the 0.30–0.60 OR range (+6.6%), where partial overlap introduces substantial appearance ambiguity. Smaller but stable improvements are also observed under both low and severe occlusion conditions. Furthermore, confusion statistics show that the total number of mutual misclassifications among visually similar behaviors decreases after introducing MFI. The number of mutual misclassifications between Eat and Stand decreases from 9 to 4 over 260 ground-truth instances. This reduction suggests that MFI helps distinguish behaviors with similar static appearances by incorporating temporal features across adjacent frames. For Lay, which contains subtle posture transitions across adjacent frames, MFI achieves comparable accuracy to STAR (86.6% vs. 87.1%, a margin of 0.5%) while reducing FLOPs. These results indicate that MFI can capture fine-grained temporal features with substantially lower computational cost.

Table 7
Detection performance across different OR ranges.

OR Range	RT-DETR (baseline)	RT-DETR+MFI
< 0.10	92.0	93.0
0.10–0.30	90.9	93.4
0.30–0.60	83.0	89.6
≥ 0.60	76.9	78.3

3.3.3. *Feature pyramid architecture comparison*

To evaluate the performance of the proposed PGPN, we conducted comparative experiments on several feature pyramid architectures, including the Adaptive Structure Feature Fusion (ASFF) [30], the Bidirectional Feature Pyramid Network (BiFPN) [31], the Scale Sequence Feature-Based Feature Pyramid Network (SSFPN) [45], and CNN-based Cross-scale Feature Fusion (CCFF) [22]. These architectures were selected to represent the major trends in feature pyramid design, encompassing adaptive feature reweighting (ASFF), bidirectional multi-scale fusion (BiFPN), sequential scale integration (SSFPN), and convolutional cross-scale feature interaction (CCFF). Detailed experimental results are presented in Table 8.

Comprehensive evaluations demonstrate that PGPN consistently achieves the highest detection accuracy across all four behaviors, with AP_{50} scores of 94.0%, 86.4%, 80.3%, and 52.6% for Eat, Lay, Stand, and Drink, respectively. In addition, PGPN maintains excellent computational efficiency, requiring only 120G FLOPs, running at 51 FPS, and containing 39M parameters. Compared with other feature pyramid modules, PGPN achieves a more favorable balance between accuracy

Table 8
Performance comparison of different feature pyramid.

Methods	AP_{50} (%)				Params (M)	$FPS_{bs=1}$	FLOPs (G)
	Eat	Lay	Stand	Drink			
+ASFF	92.3	83.9	74.9	52.1	40	50	134
+BiFPN	91.8	83.5	79.1	49.7	40	51	122
+SSFPN	91.5	84.3	78.1	51.6	49	33	182
+CCFF	93.2	83.4	77.3	50.9	44	49	137
+PGPN (our)	**94.0**	**86.4**	**80.3**	**52.6**	**39**	**51**	**120**

Table 6
Performance comparison of different spatiotemporal feature extraction methods.

Methods	AP_{50} (%)				Params(M)	$FPS_{bs=1}$	FLOPs(G)
	Eat	Lay	Stand	Drink			
+C3D	91.4	77.0	70.4	45.4	121	33	208
+X3D	92.4	85.3	76.9	51.9	**66**	31	216
+VideoSwin	94.0	84.6	81.5	52.3	177	22	275
+STAR	**96.6**	**87.1**	**85.1**	51.3	82	10	555
+MFI(our)	95.9	86.6	84.6	**56.3**	88	**35**	**187**

and efficiency, confirming its effectiveness in enhancing sheep behavior detection under complex farming environments.

3.4. Inference efficiency and consistency verification

DRSTAF re-parameterizes its weights after training, the data pipelines used in training and inference differ structurally. To ensure that this re-parameterization does not alter the functional behavior of the model, we conducted an consistency validation using the same set. This validation focuses on confirming the consistency between the two pipelines while also demonstrating that the re-parameterized inference pipeline achieves faster inference speed. The corresponding results are presented in Table 9.

The experimental results reveal two key observations: (1) AP_{50} demonstrates remarkable consistency across all behavioral categories and average deviation of merely 0.45 percentage points. For high-frequency behaviors (lay: 98.3% vs. 98.3%; stand: 97.8% vs. 97.7%; eat: 98.6 %vs. 98.5%), AP_{50} deviations remain $\leq$ 0.1%, confirming stable feature preservation. The observed deviation of 1.6% in the drink category can be attributed to its limited sample size, which impacted the consistency between the Training-mode Model and Inference-mode Model. To further analyze this deviation, we perform a feature-level comparison between the original multi-branch structure and the re-parameterized single-branch structure under identical inputs. Specifically, cosine similarity is computed between intermediate feature maps. The results show consistently high similarity (average cosine similarity is 0.989), demonstrating that the fused representation closely approximates the original feature distribution. (2) Significant efficiency gains emerge during inference, with 33% Params reduction, 16% lower FLOPs, and 34% faster $FPS_{bs=1}$, demonstrating effective model optimization.

This empirical validation confirms that our different data pipelines architecture achieves dual optimization objectives: maintaining feature consistency while realizing inference acceleration. The Params compression and computational efficiency demonstrate successful implementation of model lightweighting strategies during inference phase. Notably, these efficiency improvements are attained without compromising detection accuracy. These findings provide strong evidence that STRep-DETR can reliably maintain its learning capacity when transitioning from training to inference mode.

3.5. Visualized performance across varied lighting & densities

To further substantiate the detection robustness of STRep-DETR, we conduct qualitative comparisons against RT-DETR under various illumination and sheep-density conditions. As depicted in Fig. 9 and Fig. 10, each comparison set displays, from left to right, the original image, RT-DETR detection, and STRep-DETR detection. Yellow circles highlight critical detection differences between the two methods. The experiments encompass both daytime and nighttime scenes, each divided into high- and medium-density cases, to comprehensively evaluate model adaptability across realistic farm environments.

Table 9
Result of different data pipelines. Training-mode Model: Complete architecture with all training-time components and data pipeline. Inference-mode Model: Model with Re-parameterization techniques and partial data pipeline.

Model	AP_{50} (%)				Params (M)	$FPS_{bs=1}$	FLOPs (G)
	Eat	Lay	Stand	Drink			
Training-mode Model	98.6	98.3	97.8	70.3	88	35	187
Inference-mode Model	98.5	98.3	97.7	71.9	**59**	**47**	**157**

Fig. 9 illustrates representative detection results under natural daylight. In the high-density scene (a–c), the white circle indicates a sheep performing "eat" behavior that RT-DETR failed to detect due to severe occlusion from surrounding animals, whereas STRep-DETR successfully localized it. This demonstrates the effectiveness of our spatiotemporal representation module in preserving contextual cues for heavily occluded targets. In the medium-density scene (d–f), the white circle marks a standing sheep that RT-DETR misclassified as "drink" due to similar posture characteristics. STRep-DETR correctly identified the behavior category, validating the contribution of multi-scale feature fusion in distinguishing visually ambiguous postures.

Fig. 10 presents detection outcomes under low-light illumination. In the high-density nighttime scene (a–c), the white circle encompasses 6 problematic detections by RT-DETR: three sheep beneath the feeding fence were completely missed, and three lying sheep were misclassified as "stand" due to extended leg positions resembling standing posture under poor lighting. STRep-DETR resolved all six cases correctly, highlighting how temporal feature aggregation maintains perceptual continuity. In the medium-density nighttime scene (d–f), the white circle indicates a partially occluded sheep that RT-DETR failed to detect due to combined occlusion from both the fence and neighboring animals. STRep-DETR successfully detected this target, demonstrating robustness to compound occlusion scenarios.

Across all visual settings, STRep-DETR exhibits a clear reduction in common detection failures compared with RT-DETR, including missed detections of heavily occluded individuals, redundant bounding boxes in dense clusters, and posture-related classification inconsistencies. These visual improvements directly correspond to our design motivations: spatiotemporal representation module addresses occlusion challenges by leveraging temporal context, multi-scale feature fusion enhances discrimination of similar postures. These visual patterns are consistent with the overall improvements reported in Section 3.2, suggesting that the integration of spatiotemporal representation and multi-scale feature fusion jointly enhances robustness under challenging conditions, enabling reliable livestock behavior monitoring across diverse lighting and density scenarios.

3.6. Evaluation on public benchmark

To further assess the generalizability of STRep-DETR beyond the SheepAct dataset, we conducted additional experiments on the public CBVD-5 dataset [21]. CBVD-5 focuses on cow behavior detection and shares similar challenges with SheepAct, including frequent occlusions, varying stocking densities, and complex lighting conditions in intensive farming environments.

We compared STRep-DETR with representative models, including YOLOv11, YOLOv12, RT-DETR, and RT-DETR v2. All models were trained and evaluated under consistent hyperparameter settings to ensure fair comparison. The results are summarized in Table 10.

STRep-DETR achieves competitive detection accuracy across all behavior categories, demonstrating consistent superiority in stand, lay, and feed behaviors compared to RT-DETR. In terms of efficiency, STRep-DETR operates at 47 FPS with 157 GFLOPs, offering a favorable balance between speed and computational cost for real-time applications. The consistent performance across different livestock species suggests that the spatiotemporal modeling capabilities of STRep-DETR possess strong generalization potential beyond the specific conditions of the SheepAct dataset.

Although CBVD-5 involves a different animal species, the superiority of STRep-DETR suggests that our method learns robust spatiotemporal representations that generalize well across diverse intensive farming scenarios. This cross-dataset validation addresses the concern about overfitting to the self-constructed dataset and highlights the method's generalization potential.

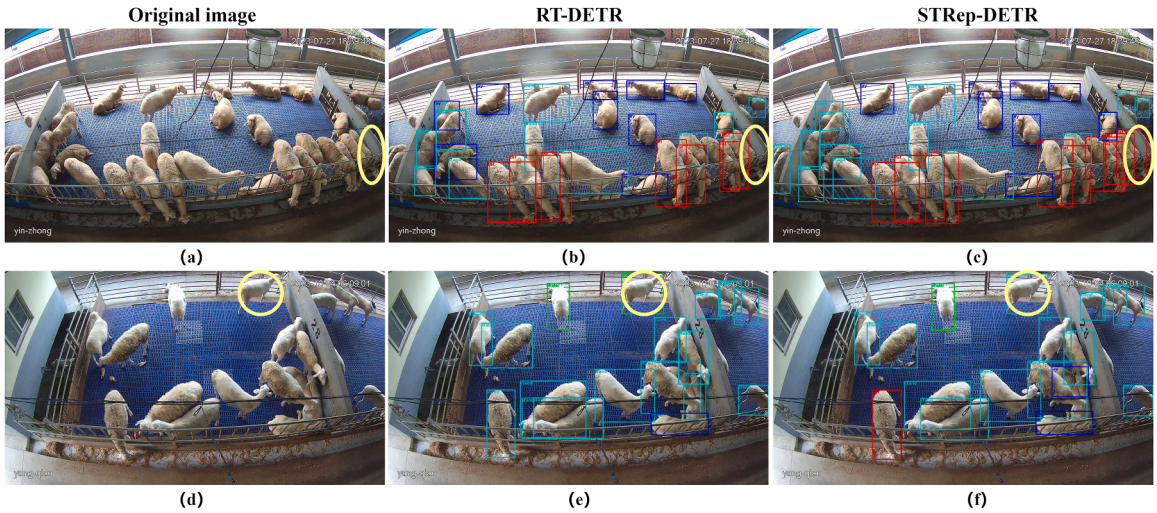


Fig. 9. Detection results under daytime conditions. From left to right: (a)(d) Original image, (b)(e) RT-DETR results, (c)(f) STRep-DETR results. The top row (a–c) corresponds to high-density daytime scenarios, and the bottom row (d–f) to medium-density ones.

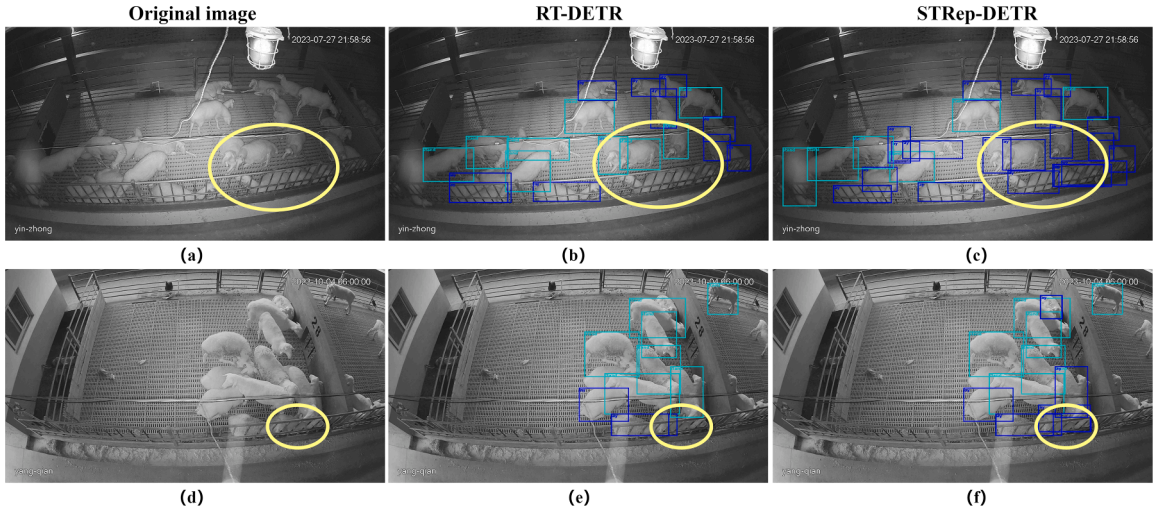


Fig. 10. Detection results under nighttime conditions. From left to right: (a)(d) Original image, (b)(e) RT-DETR results, (c)(f) STRep-DETR results. The top row (a–c) corresponds to high-density nighttime scenarios, and the bottom row (d–f) to medium-density ones.

Table 10
Performance comparison on CBVD-5 dataset.

Model	AP_{50} (%)					Params(M)	$FPS_{bs=1}$	FLOPs(G)
	stand	lay	Feed	drink	rumination			
YOLOv11 X	75.5	67.0	78.4	46.5	35.5	56.9	29	194.9
YOLOv12 X	70.7	62.1	75.0	48.7	38.7	59.1	31	199.0
RT-DETR	70.2	62.6	77.4	49.8	38.6	**44**	**49**	**137**
RT-DETRv2	73.1	63.6	78.0	49.7	38.3	44	49	137
STRep-DETR(our)	**76.9**	**67.9**	**80.3**	**51.2**	**39.3**	59	47	157

4. Discussion

To address the challenges of robust and efficient sheep behavior monitoring under medium-to-high stocking densities and dynamic lighting conditions, this study proposes STRep-DETR, an end-to-end spatiotemporal detection transformer that integrates MFI for temporal context modeling, DRSTAF for training–inference decoupling, and PGPN for lightweight multi-scale fusion. Experiments conducted on the newly constructed SheepAct dataset show that STRep-DETR achieves state-of-the-art performance while maintaining real-time inference capability, demonstrating its practical potential for intelligent livestock farming applications. The model attains 97.7% AP_{50} for eat, 90.1% for lay, 85.7% for stand, and 58.7% for drink, outperforming mainstream detectors such as YOLOs, and DETRs, particularly in high-density and nighttime environments.

The ablation results further demonstrate that each proposed component contributes distinct benefits: MFI enhances temporal continuity and improves stand detection by 8.4%, DRSTAF strengthens spatiotemporal alignment while reducing inference parameters from 88M to 59M and increasing drink accuracy by 7.8%, and PGPN provides

a favorable balance between representation quality and efficiency by reducing FLOPs by 12.4% over CCFF and improving lay accuracy by 3.0%. When combined, these modules yield synergistic gains, enabling STRep-DETR to reach the highest accuracy across all behaviors while maintaining 47 FPS and 157 GFLOPs. Visual comparisons (Fig. 9–10) confirm that the model generates tighter and more reliable bounding boxes than RT-DETR in crowded or low-contrast scenes due to its stronger suppression of background clutter and more stable spatiotemporal attention.

Compared with recent livestock behavior detection methods, STRep-DETR shows clear advantages. The approach in [18] constructs a skeletal representation framework by combining YOLOv5, DeepSORT, and AlphaPose to model spatiotemporal graphs for sheep behavior recognition; however, this multi-stage pipeline requires pose estimation and tracking as preprocessing steps, which increases complexity and may propagate errors across stages. Similarly, In [19], a Two-Stream Convolutional Network is adopted to jointly model RGB appearance and optical-flow motion cues for pig behavior recognition. Although the dual-stream design effectively captures temporal dynamics, it relies heavily on precomputed optical flow, which introduces substantial computational overhead and makes real-time inference difficult in resource-limited farm environments. In contrast, STRep-DETR achieves high AP_{50} performance with only 157 GFLOPs and, crucially, operates in a single-stage, end-to-end manner, eliminating error propagation from behavior detection.

The proposed method still faces two key limitations. The detection accuracy for drink remains notably lower than other behaviors due to its low occurrence frequency, which causes severe data imbalance, and because the sheep's head often blocks the small water trough, leading to frequent confusion with stand. Even with spatiotemporal modeling, these visually similar postures remain difficult to separate. In addition, the SheepAct dataset, although collected under realistic farm conditions, contains only four behaviors from a single commercial farm, which constrains the model's generalization capability. To address these issues, future work will incorporate multi-sensor data such as depth or thermal imaging to mitigate occlusion during drinking, expand SheepAct to include more behaviors, diverse farms, and multiple breeds for improved cross-domain robustness, and explore integrating STRep-DETR into a closed-loop farm management system that enables real-time behavioral alerts and automated responses to support fully autonomous precision livestock farming. In addition, future work will investigate deployment and performance evaluation on real edge devices to furth-er validate the model's applicability in resource-constrained environments.

5. Conclusion

This study proposes STRep-DETR, a spatiotemporal transformer designed for robust and efficient detection of sheep behaviors in intensive livestock farms under medium-to-high stocking densities and dynamic lighting conditions. The following conclusions are drawn from this research:

(1) Integrating MFI, DRSTAF, and PGPN into RT-DETR significantly improves detection accuracy, achieving 85.7% for stand, 89.3% for lay, 98.3% for eat, and 56.2% for drink on the SheepAct dataset.
(2) Ablation and comparison experiments confirm that each module enhances modeling while maintaining efficiency. STRep-DETR outperforms state-of-the-art detectors including YOLOs and DETRs, with superior accuracy under high density and low-light conditions.
(3) STRep-DETR runs at 47 FPS with only 59 million parameters and 157 GFLOPs, demonstrating favorable computational efficiency through re-parameterization, which simplifies the model structure during inference and reduces computational overhead.

The STRep-DETR model proposed in this study addresses key challenges in dense livestock behavior monitoring. It not only meets the operational requirements for automated welfare assessment but also provides a scalable method for behavior-aware vision systems in other intensive livestock settings, thereby advancing the development of intelligent, data-driven precision livestock farming.

Ethics statement

All Institutional and National Guidelines for the care and use of animals (fisheries) were followed. The experimental protocols were approved by the Experimental Animal Care and Use Committee of the Institute of Animal Sciences, China Agricultural University (approval number AW82603202-1-1).

CRediT authorship contribution statement

Zhuoyan Hou: Writing – original draft, Validation, Software, Methodology. **Wenjing Li:** Visualization, Validation, Software, Formal analysis. **Hongrui Chang:** Software, Formal analysis, Data curation. **Suoxiang Zhang:** Visualization, Validation, Resources. **Xiaxi Li:** Visualization, Software, Resources, Project administration. **Guoxin Wu:** Visualization, Formal analysis. **Ronghua Ji:** Writing – review & editing, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (No. 2022YFD1301104) and the High-performance Computing Platform of China Agricultural University.

Data availability

https://github.com/CAU-713/sheepAct (The dataset has been publicly archived and can be accessed via the URL)

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